



Tidal Prediction Checksheet

Station: 66840 Apia

Type: Secondary

Prediction Year: 2026

Executive Summary

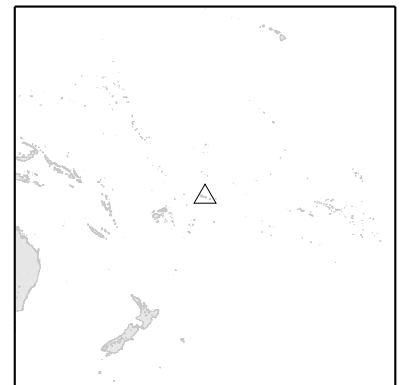
Tidal Analysis

Analysis ID: 66840_2024_tideA

Analysis period: 1993–2023

Reference station: Not registered

Given and related constituents: No



Tide Prediction

Prediction year: 2026

Prediction datum: TGZ

Datum adjustment: 0.000

MSL adjustment: 2.7 mm/yr

Long-period tide replacement: Yes

Minimum high/low water interval: 1.0 hours

Extra tides: No

Comments

Long period tides [Sa Ssa] from 1993–2023 analysis at Apia

Zero frequency level trend adjustment for 2026 predictions [18.0 years at 2.7 mm/yr] = 0.0486

Basis of 2026 Predictions – Tidal Analysis Details

Analysis epoch start: 1993-02-27
Analysis epoch end: 2023-12-31
Analysis epoch years: 30.84
Days of data: 10920
Years of data: 29.90
Data completeness index: 0.97
Number standard constituents: 112
Number other constituents: 0
Time polynomial degree: 0
Sample correlation coefficient: 0.9318
Standard deviation of residuals: 0.1166
Analysis zero frequency level (A0): 0.8393

Basis of 2026 Predictions – Tidal Constants

Timezone: UTC+13:00

Prediction zero frequency level (Z0): 0.888

NAME	FREQ(deg/hr)	H(m)	g(deg)	NAME	FREQ(deg/hr)	H(m)	g(deg)
Sa	0.0410686	0.0079	116.7	2MS6	87.9682084	0.0007	62.4
Ssa	0.0821373	0.0021	31.1	2MK6	88.0503457	0.0002	64.0
Mm	0.5443747	0.0063	20.5	2SM6	88.9841042	0.0002	125.9
Msf	1.0158958	0.0008	358.4	MSK6	89.0662415	0.0001	130.6
Mf	1.0980331	0.0095	34.8	2MN2S2	26.4079379	0.0001	67.1
2Q1	12.8542862	0.0008	174.4	3M(SK)2	26.8701753	0.0004	286.0
SIGMA1	12.9271398	0.0008	186.9	3M2S2	26.9523127	0.0003	87.2
Q1	13.3986609	0.0055	213.4	MNK2S2	27.5059710	0.0002	284.5
RH01	13.4715145	0.0011	206.9	SNK2	28.3575922	0.0007	180.5
O1	13.9430356	0.0316	227.9	2SK2	29.9178627	0.0007	241.9
MP1	14.0251729	0.0009	338.9	MQ3	42.3827651	0.0001	304.3
M1	14.4920521	0.0014	240.1	2MP3	43.0092770	0.0002	26.0
CHI1	14.5695476	0.0004	249.4	2MQ3	44.5695475	0.0000	54.4
PI1	14.9178647	0.0007	270.4	3MK4	56.8701754	0.0001	357.8
P1	14.9589314	0.0128	249.3	3MS4	56.9523127	0.0002	30.3
S1	15.0000000	0.0045	100.5	2MSK4	57.8860711	0.0002	169.5
K1	15.0410686	0.0423	257.7	3MK5	71.9112441	0.0002	60.3
PSI1	15.0821353	0.0010	289.7	M5	72.4602605	0.0001	237.2
PHI1	15.1232059	0.0010	322.2	3M05	73.0092771	0.0001	186.7
THETA1	15.5125897	0.0004	282.3	2(MN)S6	84.8476674	0.0001	240.4
J1	15.5854433	0.0023	269.4	3MNS6	85.3920422	0.0000	198.3
S01	16.0569644	0.0007	226.8	4MK6	85.8542795	0.0000	139.4
O01	16.1391017	0.0007	353.0	4MS6	85.9364168	0.0001	100.2
OQ2	27.3416964	0.0003	318.6	2MSNK6	86.3258006	0.0001	121.7
MNS2	27.4238337	0.0055	121.8	2MNU6	86.4807915	0.0001	332.2
2N2	27.8953548	0.0167	138.7	3MSK6	86.8701754	0.0000	237.2
MEU2	27.9682084	0.0197	141.0	4MN6	87.4966873	0.0001	243.1
N2	28.4397295	0.1064	163.6	3MSN6	88.5125832	0.0002	272.0
NEU2	28.5125831	0.0199	168.6	MKL6	88.5947204	0.0001	68.2
OP2	28.9019669	0.0013	38.2	2MN8	114.8476674	0.0000	277.6
M2	28.9841042	0.3897	182.3	3MN8	115.3920422	0.0001	303.6
MKS2	29.0662415	0.0005	130.5	M8	115.9364168	0.0000	344.7
LAMDA2	29.4556253	0.0019	220.5	2MSN8	116.4079379	0.0001	25.6
L2	29.5284789	0.0061	193.2	3MS8	116.9523127	0.0001	61.2
T2	29.9589333	0.0072	184.3	3MK8	117.0344500	0.0000	27.5
S2	30.0000000	0.0949	183.2	MSNK8	117.5059710	0.0000	156.2
R2	30.0410667	0.0007	160.7	2MS8	117.9682084	0.0001	162.3
K2	30.0821373	0.0262	176.4	2MSK8	118.0503457	0.0000	178.9
MSN2	30.5443747	0.0006	216.4	4MS10	145.9364168	0.0000	185.8
KJ2	30.6265120	0.0017	351.6	3M2S10	146.9523127	0.0000	311.1
2SM2	31.0158958	0.0004	246.6	4MSN12	174.3761463	0.0000	219.0
M03	42.9271398	0.0001	23.0	5MS12	174.9205210	0.0000	181.5
M3	43.4761563	0.0039	224.5	4M2S12	175.9364168	0.0000	28.0
S03	43.9430356	0.0003	173.2	SP3	44.9589314	0.0021	275.9
MK3	44.0251729	0.0006	277.3	S3	45.0000000	0.0009	146.8
SK3	45.0410686	0.0022	277.4	MA2	28.9430375	0.0012	235.6
MN4	57.4238337	0.0008	357.5	Ma2	29.0251709	0.0019	153.3
M4	57.9682084	0.0029	70.2	MSV2	30.4715211	0.0000	117.3
SN4	58.4397295	0.0003	85.0	SKM2	31.0980331	0.0003	206.1
MS4	58.9841042	0.0004	162.2	2MNS4	56.4079380	0.0000	200.2
MK4	59.0662415	0.0002	143.0	MV4	57.4966873	0.0002	353.4
S4	60.0000000	0.0011	114.9	3MN4	58.5125832	0.0002	327.8
SK4	60.0821373	0.0003	335.8	2MSN4	59.5284789	0.0001	12.4
2MN6	86.4079380	0.0002	322.4	NA2	28.3986628	0.0008	10.7
M6	86.9523127	0.0005	20.3	MS05	72.9271398	0.0001	246.9
MSN6	87.4238337	0.0002	34.6				

Basis of 2026 Predictions – Comparison of major tidal constituents

Comparison of tidal constituents

Set 1: 66840_2024_tideA (1993–2023 Analysis)

Set 2: ws1993–2022 (1993–2022 Analysis)

Constituents	Set 1		Set 2		Comparison	
Name	H	g(deg)	H	g(deg)	Ratio	Diff(mins)
Sa	0.0079	116.6890	0.0066	151.9524	1.20	-51518.8
Ssa	0.0021	31.1070	0.0020	78.3811	1.05	-34533.0
Q1	0.0055	213.4166	0.0055	213.2767	1.00	0.6
O1	0.0316	227.8886	0.0317	228.0264	1.00	-0.6
P1	0.0128	249.2870	0.0129	249.3899	0.99	-0.4
K1	0.0423	257.7350	0.0424	257.6893	1.00	0.2
N2	0.1064	163.5757	0.1063	163.5710	1.00	0.0
M2	0.3897	182.3247	0.3897	182.3008	1.00	0.0
S2	0.0949	183.2162	0.0948	183.2265	1.00	-0.0
K2	0.0262	176.3981	0.0262	176.3898	1.00	0.0
A0	0.8393	0.0000	0.8821	0.0000	0.95	0.0

January 2026 Tide Predictions

L	ANTT	YYYY-MM-DD	TIME	HGT	TIME	HGT	TIME	HGT	TIME	HGT	TIME	HGT
0	66840	2026 01 01	0406	1.29	1002	0.40	1625	1.49	2255	0.31		
0	66840	2026 01 02	0502	1.32	1055	0.36	1719	1.53	2347	0.26		
0	66840	2026 01 03	0556	1.35	1147	0.34	1813	1.55				
1	66840	2026 01 04	0039	0.23	0646	1.36	1239	0.35	1905	1.54		
1	66840	2026 01 05	0129	0.25	0737	1.35	1331	0.38	1957	1.50		
1	66840	2026 01 06	0218	0.30	0827	1.32	1425	0.43	2047	1.43		
1	66840	2026 01 07	0307	0.37	0917	1.28	1519	0.50	2138	1.35		
1	66840	2026 01 08	0356	0.45	1012	1.23	1615	0.56	2230	1.26		
1	66840	2026 01 09	0446	0.53	1110	1.19	1715	0.62	2328	1.18		
1	66840	2026 01 10	0540	0.59	1211	1.17	1817	0.65				
0	66840	2026 01 11	0030	1.12	0638	0.63	1310	1.16	1925	0.66		
0	66840	2026 01 12	0135	1.08	0739	0.64	1403	1.18	2030	0.64		
0	66840	2026 01 13	0236	1.07	0833	0.63	1451	1.20	2124	0.61		
0	66840	2026 01 14	0330	1.08	0920	0.61	1534	1.22	2209	0.56		
0	66840	2026 01 15	0415	1.09	1000	0.58	1615	1.25	2247	0.51		
0	66840	2026 01 16	0454	1.12	1037	0.55	1654	1.28	2323	0.47		
0	66840	2026 01 17	0530	1.15	1113	0.52	1730	1.32	2357	0.44		
0	66840	2026 01 18	0603	1.17	1147	0.50	1807	1.34				
1	66840	2026 01 19	0030	0.42	0637	1.20	1223	0.48	1844	1.37		
1	66840	2026 01 20	0104	0.41	0711	1.23	1301	0.48	1921	1.38		
1	66840	2026 01 21	0139	0.41	0746	1.25	1342	0.48	2000	1.37		
1	66840	2026 01 22	0216	0.43	0826	1.26	1427	0.50	2043	1.35		
1	66840	2026 01 23	0257	0.45	0910	1.26	1516	0.52	2129	1.31		
1	66840	2026 01 24	0341	0.48	1001	1.25	1613	0.55	2222	1.26		
1	66840	2026 01 25	0432	0.52	1102	1.24	1716	0.56	2324	1.21		
1	66840	2026 01 26	0532	0.55	1210	1.25	1828	0.56				
0	66840	2026 01 27	0036	1.17	0642	0.56	1318	1.28	1943	0.53		
0	66840	2026 01 28	0151	1.17	0754	0.54	1423	1.33	2054	0.46		
0	66840	2026 01 29	0300	1.20	0900	0.49	1524	1.40	2155	0.39		
0	66840	2026 01 30	0401	1.25	0958	0.43	1619	1.46	2248	0.32		
0	66840	2026 01 31	0456	1.31	1049	0.38	1712	1.51	2337	0.27		

Differences from Previous Basis of Predictions (in minutes and cm)

L	ANTT	YYYY-MM-DD	TIME	HGT	TIME	HGT	TIME	HGT	TIME	HGT	TIME	HGT
0	66840	2026 01 01	0	1	0	0	0	0	0	0		
0	66840	2026 01 02	0	0	0	0	0	0	-1	1		
0	66840	2026 01 03	0	0	0	0	0	0				
1	66840	2026 01 04	0	0	0	0	0	1	0	0		
1	66840	2026 01 05	0	0	1	0	0	0	1	0		
1	66840	2026 01 06	0	1	0	0	0	0	1	0		
1	66840	2026 01 07	0	1	0	0	0	1	1	0		
1	66840	2026 01 08	0	0	1	0	0	0	0	0		
1	66840	2026 01 09	1	0	0	0	0	0	0	0		
1	66840	2026 01 10	0	0	0	0	0	0				
0	66840	2026 01 11	0	1	0	0	0	0	0	0		
0	66840	2026 01 12	0	0	0	0	0	1	0	0		
0	66840	2026 01 13	0	0	0	0	0	1	0	1		
0	66840	2026 01 14	1	1	1	0	0	0	0	0		
0	66840	2026 01 15	0	0	0	1	0	0	0	0		
0	66840	2026 01 16	0	1	0	1	1	0	0	0		
0	66840	2026 01 17	0	1	1	1	0	1	0	0		
0	66840	2026 01 18	0	0	0	1	0	0				
1	66840	2026 01 19	0	0	0	0	0	0	0	1		
1	66840	2026 01 20	0	0	0	0	0	1	0	1		
1	66840	2026 01 21	0	0	0	0	0	0	0	0		
1	66840	2026 01 22	0	1	0	1	0	1	1	0		
1	66840	2026 01 23	0	0	0	1	0	0	0	0		
1	66840	2026 01 24	0	0	0	0	0	1	1	1		
1	66840	2026 01 25	0	1	0	0	0	0	0	1		
1	66840	2026 01 26	1	1	0	0	0	0				
0	66840	2026 01 27	0	1	1	1	0	0	-1	1		
0	66840	2026 01 28	0	1	0	1	-1	0	0	0		
0	66840	2026 01 29	0	1	0	1	0	1	0	1		
0	66840	2026 01 30	-1	1	0	1	0	1	0	1		
0	66840	2026 01 31	0	1	0	1	0	1	0	0		

Extreme Tides for 2026**10 Highest Predicted Tides**

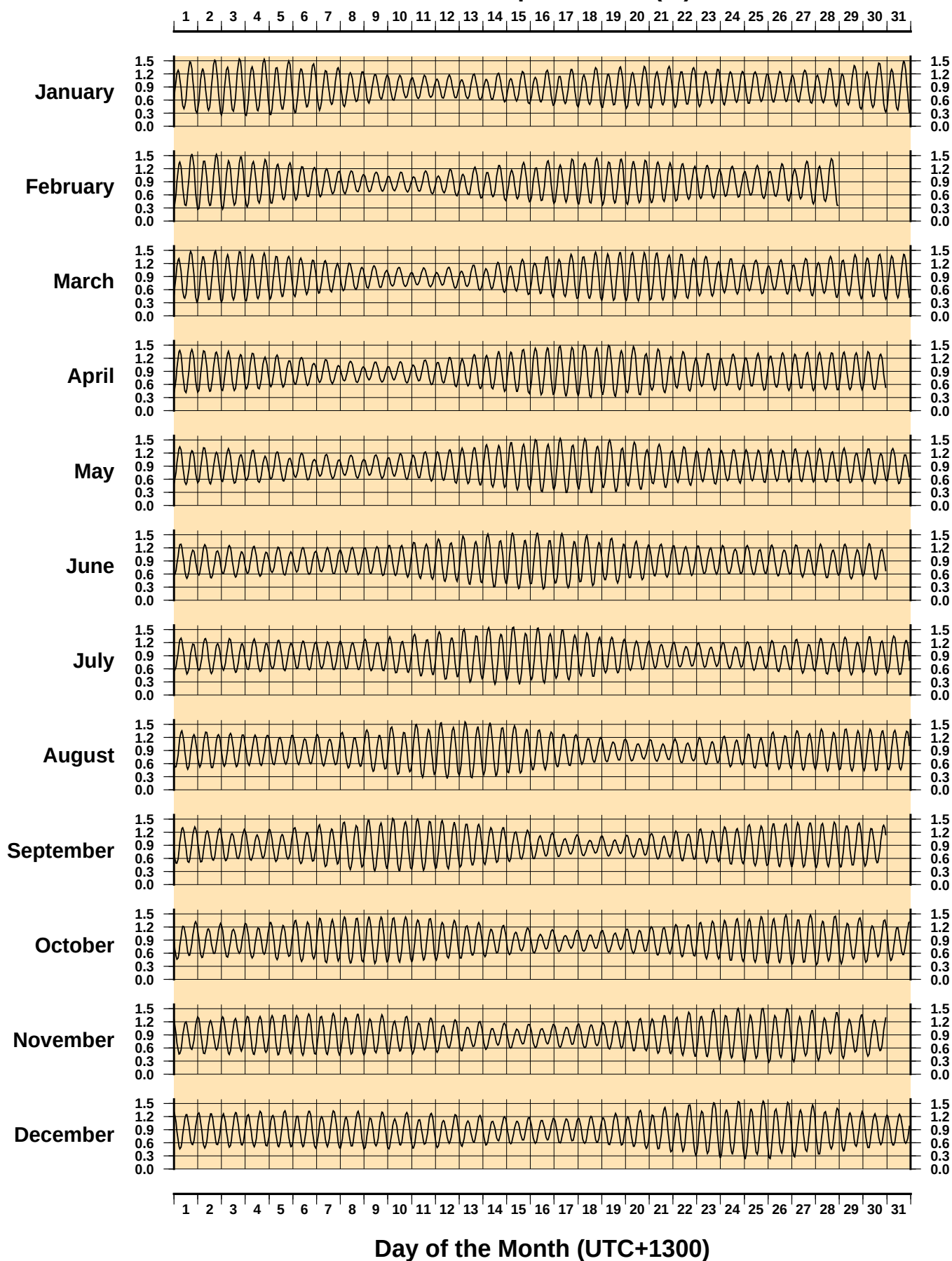
2026-07-15T06:40:00+13:00,	1.57, H
2026-12-25T19:17:00+13:00,	1.56, H
2026-12-24T18:23:00+13:00,	1.56, H
2026-08-13T06:22:00+13:00,	1.56, H
2026-08-12T05:33:00+13:00,	1.55, H
2026-07-16T07:32:00+13:00,	1.55, H
2026-07-14T05:45:00+13:00,	1.55, H
2026-06-16T06:53:00+13:00,	1.55, H
2026-01-03T18:13:00+13:00,	1.55, H
2026-06-15T05:57:00+13:00,	1.54, H

10 Lowest Predicted Tides

2026-12-25T00:49:00+13:00,	0.22, L
2026-01-04T00:39:00+13:00,	0.23, L
2026-12-26T01:43:00+13:00,	0.23, L
2026-01-05T01:29:00+13:00,	0.25, L
2026-06-16T13:18:00+13:00,	0.25, L
2026-07-14T12:11:00+13:00,	0.25, L
2026-07-15T13:01:00+13:00,	0.25, L
2026-12-23T23:57:00+13:00,	0.25, L
2026-01-02T23:47:00+13:00,	0.26, L
2026-02-02T00:22:00+13:00,	0.26, L

Apia

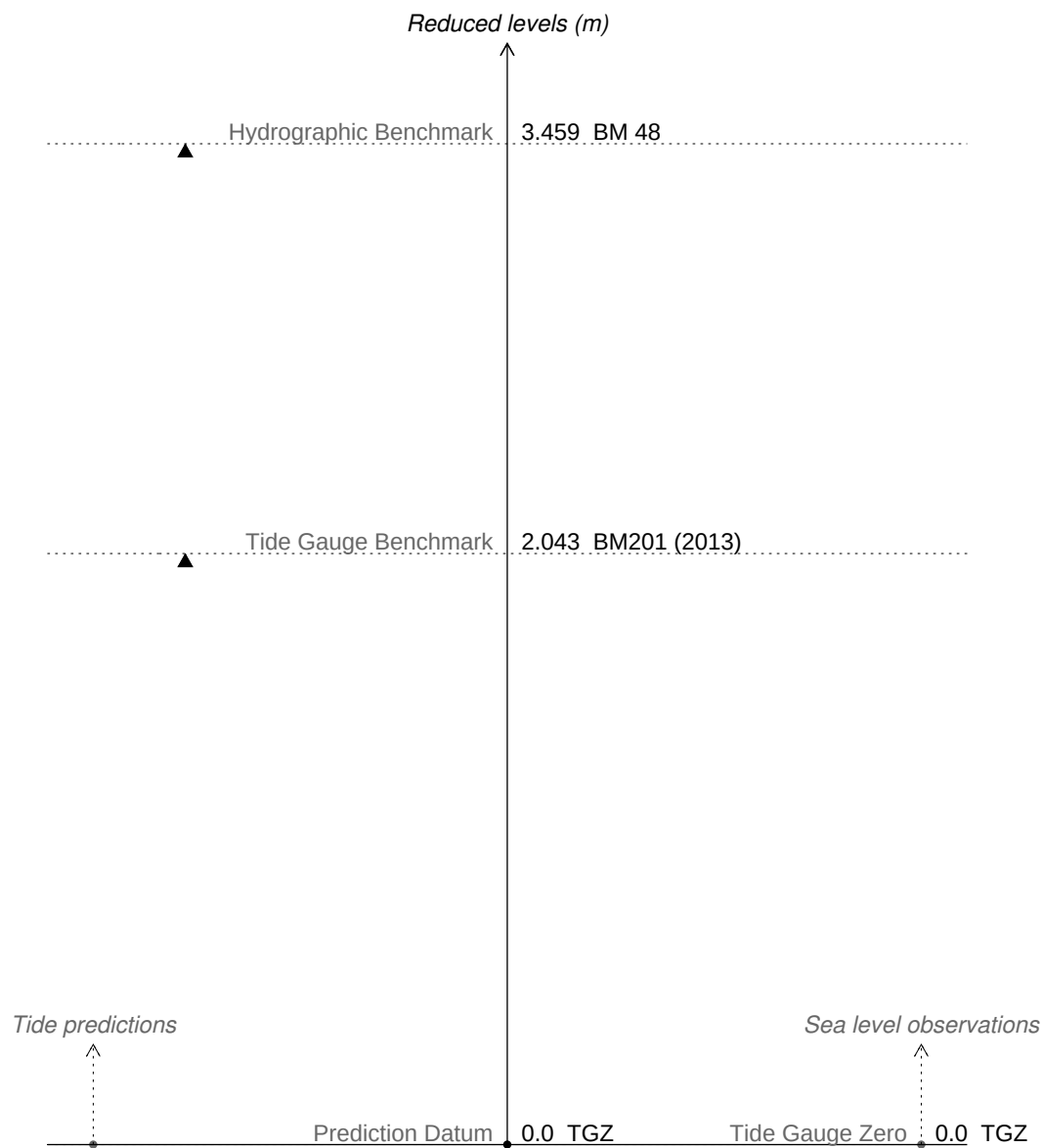
2026 Official predictions (m)



Tide Registration for Apia

SITE	
name	Apia
alt_name	SAMOA, UPOLU ISLAND - APIA
STATE	
name	
code	
COUNTRY	
name	Samoa
code	WSM
ANTT	
code	66840
alt_ref	66840.0
alt_code	
AAC	
code	INT_TP022
STN	
number	200814
array	200814 200874
ID	
abbr	ws
alt_abbr	
bank	apia
CLASSIFICATION	
type	Secondary
networks	PSLM
oper_flag	Y
gnss_array	SAMO
OPERATOR	
tide_gauge	Bureau of Meteorology
stn_ref	
data_supplier	
data_channel	CMSS
data_licence	
COORDINATES	
latitude	-13.8268
longitude	-171.7613
reference_system	
positional_uncertainty	
TIMEZONE	
utc_offset	UTC+13:00
name	Pacific/Apia
ANALYSIS	
epoch_start	1993-02-01T00:00:00Z
epoch_end	
reference_port_antt	
reference_port_name	
flags	Z0trend
PREDICTION_DATUM	
reduced_level	0.0
name	TGZ
uncertainty	
TIDE_GAUGE_ZERO	
reduced_level	0.0
name	TGZ
uncertainty	
HYDROGRAPHIC_BENCHMARK	
reduced_level	3.459
name	BM 48
uncertainty	
TIDE_GAUGE_BENCHMARK	
reduced_level	2.043
name	BM201 (2013)
uncertainty	
LAND SURVEY DATUM	
reduced_level	
name	
uncertainty	
ELLIPSOID DATUM	
reduced_level	
name	
uncertainty	
NOTES	
note1	None
note2	
REVISION	
status	Revised
date	2025-02-25T11:31:58Z

Apia
Latitude: -13.8268
Longitude: -171.7613
ANTT: 66840
AAC: INT_TP022
Revision: Revised 2025-02-25



Notes
Reduced Levels: All heights are in metres relative to prediction datum.
Prediction Datum: Applicable to tide predictions issued by the Bureau of Meteorology tides@bom.gov.au.
Scope: Not intended to replace primary and additional information available from hydrographic, land or port authorities.